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A systematic evaluation of data preprocessing and model optimization for machine learning algorithms: Using sphalerite trace element data as an example

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Abstract: Mineral chemistry combined with machine learning (ML) algorithms has been widely used in ore genesis and mineral exploration. However, a systematic investigation on the effect of data preprocessing and optimal parameters on the performance of ML methods was not enough. Based on a real unbalanced dataset with

4,312 trace element data of sphalerite from different types of deposits, this study investigated the classification performance of eight ML algorithms, including unsupervised principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE), and supervised linear discriminant analysis (LDA), partial least squares-discriminant analysis (PLS-DA), random forest (RF), support vector machine (SVM), and extremely greedy tree boosting (XGBoost). Different preprocessing methods, including missing value imputation, data transformation, and feature selection, were evaluated. For skewed distribution data, the study recommends using KNN, Mode, and Median imputation methods to reduce bias. The PCA, t-SNE, LDA, PLS-DA, and SVM algorithms perform well on data transformed by CLR or logarithmic methods, while RF and XGBoost algorithms also show good classification performance on untransformed data, especially XGBoost shows the best performance on data without imputation. Using the Recursive Feature Elimination (RFE) feature selection with the feature importance, the key features with the most discriminative capabilities can be screened out. This research not only optimizes mineral data processing workflows but also provides important support for improving the precision and accuracy of ML models.

Keywords: Machine learning; Data preprocessing; Feature selection; Mineral chemistry; Classification

1. Introduction

The advent of in-situ analysis techniques has led to an explosive growth in mineral chemistry data. With the rapid development of artificial intelligence, machine learning (ML) has provided efficient data analysis methods for the mineral deposit field (Xiong and Zuo, 2020). The integration of ML algorithms with mineral chemistry has been widely used in understanding ore genesis and assisting mineral exploration. For example, Makvandi et al. (2019) used principal component analysis (PCA) algorithms to analyze mineral liberation analysis data of iron oxides for provenance discrimination. Partial least squares-discriminant analysis (PLS-DA) was used to discriminate rock and ore deposit types based on trace element data of apatite, magnetite, molybdenite, and sphalerite (Huang et al., 2019a, b, 2024; Xie et al., 2022; Tan et al., 2023a, b). Gregory et al. (2019) developed a classification model for pyrite trace element data using the random forest (RF) algorithm, accurately distinguishing types of ore deposits. Huang et al. (2022) used seven machine learning algorithms to estimate $\text{Fe}^{3+}/\Sigma\text{Fe}$ in clinopyroxene based on electron microprobe data. Zhu et al. (2023) used RF, Extremely Greedy tree Boosting (XGBoost), and light gradient boosting machine (LightGBM) algorithms to classify quartz deposits and identified key geochemical indicators for distinguishing different types of quartz deposits. Luo et al. (2024) used supervised machine learning algorithms, including support vector machine (SVM), RF, artificial neural network (ANN), and XGBoost, combined with zircon chemical data to identify the tectonic environments of porphyry copper deposits. Meng et al. (2024) used various machine learning algorithms such as PCA, PLS-DA, RF and SVM to discriminate deposit types based on trace elements in sphalerite. These studies have provided good examples for the combination of ML with mineral chemistry in classification issues. However, a systematic evaluation of data preprocessing and algorithm execution on the model performance was lacking.

Some studies discussed the effect of data preprocessing on the ML model performance. For example, in the data cleaning step, Bédard et al. (2022) removed magnetite data of unknown origin, abnormal data that didn't meet specified element thresholds, and elements with missing or deleted values exceeding 40% of the data. The `IrEM()` function from the `zCompositions` R package was used to fill in missing values

and it was proved that centered log-ratio (Clr) data transformation had minimal impact on the prediction accuracy of random forest models. Li et al. (2023) used the quantile regression imputation of left-censored data (QRILC) method to address missing values in the collected trace element data of sphalerite, and performed logarithmic transformations on all trace element data. These two preprocessing steps significantly enhanced the performance of the dataset in algorithm applications.

In machine learning, selecting and tuning the appropriate model hyperparameters is a key step to enhance model performance. Hyperparameters, unlike model parameters, are not learned during the training process but need to be set by researchers before training. For example, in the random forest algorithm, the depth of the trees and the number of trees are two critical hyperparameters. The selection of these hyperparameters directly affects the performance and quality of the algorithm's output. Due to the diversity of data features, different datasets may have different optimal values for the same algorithm's hyperparameters (Probst et al., 2019). Previous studies have developed various optimization methods to automatically adjust these parameters to achieve the best learning outcomes, including grid search, random search, Bayesian optimization (Shahriari et al., 2015; Liashchynskyi and Liashchynskyi, 2019). Each of these methods may perform well under specific circumstances. Therefore, it is necessary to systematically evaluate the effect of the data preprocessing, feature selection and hyperparameter tuning on the model performance.

This study used data from over 100 global lead-zinc deposits collected by Meng et al. (2024). The dataset includes 12 elements: Fe, Pb, Mn, Co, Cu, Ga, Ge, Ag, Cd, In, Sn, Sb, as well as five types of deposits: Mississippi Valley-type (MVT), volcanogenic massive sulfide (VMS), sedimentary exhalative (SEDEX), Epithermal, and Skarn. In the machine learning section, several widely used algorithms were selected, including PCA, t-distributed stochastic neighbor embedding (t-SNE), linear discriminant analysis (LDA), PLS-DA, RF, SVM, and XGBoost. The research aims to delve into the importance of data preprocessing in machine learning and its potential impact on analysis results. This study not only analyzed the impact of each preprocessing technique on individual algorithms but also examined their performance in multi-algorithm ensembles. This approach helps reveal the relative advantages of different preprocessing strategies within an integrated machine learning framework, providing more precise and effective guidance for geoscientific data processing.

2. Dataset and methods

Six missing value handling methods were applied to the original dataset, including five conventional imputation techniques and one additional approach of retaining missing entries for algorithms with native support (Zhang et al., 2024). For each imputed dataset, we applied seven data transformation methods and conducted

preliminary evaluation using unoptimized algorithms to identify the optimal missing value imputation and transformation combinations for each algorithm. Subsequently, within each algorithm, Recursive Feature Elimination (RFE) was used for feature selection and combined cross-validation and grid search was used to optimize algorithm parameters. All processes were implemented using the Python language, with the primary library being scikit-learn. The workflow was shown in Fig. 1.

2.1. Dataset

This study uses a dataset including 4312 sphalerite LA-ICP-MS trace element analyses from over 100 globally lead-zinc deposits (Meng et al., 2024). This unbalanced dataset represents practical miner chemistry dataset elsewhere. Features including 12 specific elements, Fe (0.2% missing value), Pb (1.7%), Mn (0.5%), Co (6.5%), Cu (0.1%), Ga (5.9%), Ge (4.8%), Ag (0.075%), Cd (0.2%), In (3.5%), Sn (3.5%), and Sb (7.4%), were mainly chosen due to their high detection rates. The dataset covers five main types of Pb-Zn deposits, including MVT, VMS, SEDEX, Epithermal, and Skarn. Detailed information and data are compiled in the MinerGeochem database (<https://minergeochem.com>).

2.2. Missing data imputation

When dealing with datasets containing missing values, adopting appropriate imputation methods is crucial for subsequent data analysis and model training (Acock, 2005). Here are five commonly used missing value imputation methods, along with their basic principles and applicable scenarios:

K-Nearest Neighbors (KNN) imputation estimates missing values by finding K nearest neighbors (most similar records) of the sample with missing values in the feature space (Peterson, 2009). It is suitable for cases where strong correlations exist between data features. This method is particularly appropriate for both categorical and numerical data, and can well preserve the original data distribution.

Mean imputation replaces missing values with the average of the corresponding feature. It calculates the mean of non-missing values for each feature containing missing values and uses this mean to fill all missing values (Donders et al., 2006). This method is suitable when data features follow approximately normal distribution with insignificant skewness. While computationally simple, it may affect data variance.

Random Forest imputation predicts missing values through the random forest algorithm based on other features. It can handle large amounts of data and extract complex relationships between features (Shah et al., 2014). It is suitable for datasets

with complex feature relationships or random missing patterns. While highly accurate, it has relatively high computational costs.

Similar to mean imputation, median imputation uses the median (50th percentile) of the feature to replace missing values (Donders et al., 2006). It is suitable for skewed distributions as the median is less sensitive to extreme values.

Mode imputation uses the mode (most frequent value) of the feature to replace missing values (Donders et al., 2006). It is suitable for imputing missing values in categorical features, where the mode represents the most likely category.

2.3. Data transformation

Data transformation plays a crucial role in geochemical data processing, primarily due to several characteristics of geochemical data: First, most geochemical data do not follow normal distribution but may exhibit skewed or multimodal distributions. Second, geochemical data often contains extreme values or outliers, and data transformation can effectively reduce their impact on overall data analysis. Furthermore, geochemical studies frequently involve multivariate analyses, such as principal component analysis and cluster analysis, which typically require normally distributed data with equal variance among variables (Ferketich and Verran, 1994). When applying machine learning algorithms to process geochemical data, appropriate data transformation can optimize algorithm performance (Sharmila et al., 2024). Finally, data transformation can improve data visualization, making patterns and trends more apparent. In this study, to effectively process and analyze data, we used various data transformation methods, each with its unique advantages and application scenarios. The specific methods include are simply introduced here.

Centered Log Ratio (CLR) transformation is Primarily used for compositional data, such as elemental proportions in geochemical data. It converts all variable values to logarithmic form and then subtracts the mean of all logarithmic values from each logarithmic value (Muriithi, 2015). This transformation addresses the non-closure problem in data, making variables comparable.

Decimal Scaling normalizes data by moving the decimal point of data points. Each data point is divided by a power of 10 based on the maximum absolute value, ensuring the maximum absolute value is less than 1 (Sinsomboonthong, 2022).

Log Transformation was used to reduce data skewness, making it closer to normal distribution and reducing the impact of extreme values. Values are replaced with their natural logarithms (Egozcue et al., 2003). This method is typically suitable for data with exponential growth or large ranges.

MaxAbs Scaling technique scales each feature by its maximum absolute value, transforming data to the range $[-1, 1]$. Each data point is divided by the maximum absolute value of that feature (Ahsan et al., 2021).

Mean Centering eliminates bias in average values between different data dimensions and subtracts the mean of each feature from its values, repositioning the data center to the origin (Iacobucci et al., 2016).

Min-Max Scaling technique scales data to a specified range, typically 0 to 1. Uses the formula $X' = a\left(\frac{X - X_{min}}{X_{max} - X_{min}}\right) + (-a)$ for data scaling (Ahsan et al., 2021).

Robust Scaling reduces the influence of outliers and improves statistical robustness of the dataset. Uses interquartile range for scaling data, centering by the median and normalizing by the interquartile range (IQR) (Ahsan et al., 2021).

Z-score Normalization method standardizes data to have a mean of 0 and standard deviation of 1. Normalizes data by subtracting the mean of each feature and dividing by its standard deviation (Cheadle et al., 2003).

2.4. ML algorithms

This study primarily selected seven algorithms: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), Partial Least Squares Discriminant Analysis (PLS-DA), Random Forest (RF), Support Vector Machine (SVM), t-Distributed Stochastic Neighbor Embedding (t-SNE), and eXtreme Gradient Boosting (XGBoost) (Fig. 2). Each algorithm has its unique advantages and application scenarios.

The PCA is a statistical method that transforms a set of potentially correlated variables into a set of linearly uncorrelated variables through orthogonal transformation (Abdi and Williams, 2010). These new variables are called principal components. The number of principal components is usually less than the number of original variables while preserving as much variance as possible from the original data. However, PCA is very sensitive to outliers in the data. It is widely used in data exploration, preprocessing, and dimensionality reduction, particularly suitable for handling the internal structure of multivariate data. In terms of principal component adjustment, the PCA algorithm can optimize the data dimensionality reduction process by selecting different numbers of principal components (Serneels and Verdonck, 2008). Researchers can flexibly adjust the number of principal components based on cumulative explained variance proportion or specific thresholds, achieving data compression while preserving critical information. This adjustment strategy enables PCA to be widely applied in data exploration, preprocessing, and dimensionality reduction, particularly suitable for handling the internal structure of multivariate data.

The LDA is a supervised learning method used for data classification and dimensionality reduction. It aims to project data onto a lower-dimensional space while maintaining maximum separation between different classes (Balakrishnama and Ganapathiraju, 1998). This algorithm assumes that data in each category approximately follows a Gaussian distribution and that different categories share the same covariance matrix (Fu, 2023). It achieves its goal by maximizing between-class scatter (distance between categories) and minimizing within-class scatter (distance within categories). The LDA algorithm is commonly used in pattern recognition, machine learning, and feature extraction in data compression. In terms of principal component adjustment, LDA can control the number of principal components after dimensionality reduction by adjusting the number of projection axes, thereby achieving effective data compression and feature extraction while maintaining class discriminability.

The PLS-DA is an extension of PLS regression and a supervised dimensionality reduction technique used for classification problems. PLS-DA attempts to find components that explain variations in both predictor and response variables (Brereton and Lloyd, 2014). This method is particularly suitable for problems where the number of predictor variables far exceeds the number of samples, and is widely applied in chemometrics, bioinformatics, and other fields. In terms of principal component adjustment, the PLS-DA algorithm can optimize model performance by flexibly selecting and adjusting the number of principal components. Researchers can dynamically modify the number of principal components based on cross-validation, prediction error, or specific evaluation metrics, thereby improving classification accuracy while maintaining the model's interpretability. This adjustment strategy enables PLS-DA to better handle high-dimensional data, particularly in scenarios where the number of predictor variables significantly exceeds the number of samples. The method finds widespread application in chemometrics, bioinformatics, and other fields, effectively addressing complex classification problems.

The RF is an ensemble learning technique containing multiple decision trees that introduces randomness during training to enhance model generalization (Ho, 1995). It improves predictive performance by constructing multiple decision trees and aggregating their results. This method demonstrates robust performance across multiple domains, especially when handling datasets with high-dimensional feature spaces. Regarding hyperparameter tuning in random forests, researchers can flexibly adjust multiple key parameters to optimize model performance. The primary adjustable hyperparameters include the number of trees, maximum depth of each tree, minimum number of samples required for node splitting, and the proportion of randomly selected features (Probst et al., 2019). Through systematic parameter search and cross-validation, researchers can identify the optimal hyperparameter combination, significantly enhancing the model's predictive accuracy and generalization capability in high-dimensional feature spaces. This fine-tuning enables random forests to demonstrate

robust performance across multiple domains, particularly when handling complex, high-dimensional datasets.

The SVM is a supervised learning model used for classification and regression problems. It constructs one or more hyperplanes in the data space to separate different categories as much as possible. In high-dimensional space, SVM attempts to find hyperplanes that maximize the margin between classes, showing good accuracy, especially when class distributions are complex or data dimensionality is high (Cortes and Vapnik, 1995). In SVM hyperparameter tuning, researchers can optimize model performance by adjusting key parameters. The primary adjustable hyperparameters include kernel function types (linear, polynomial, radial basis function, etc.), regularization parameter C (controlling model complexity and preventing overfitting), kernel-specific parameters (such as the γ parameter for radial basis function), parameters of different kernel functions, and sample weights. Through systematic parameter search and cross-validation, researchers can determine the optimal hyperparameter combination, significantly enhancing model performance in complex classification problems. This fine-tuning enables SVM to better adapt to different data distributions, especially when handling high-dimensional and non-linearly separable datasets.

The t-SNE is a statistical method for visualizing high-dimensional data. Its re-expresses high-dimensional data in lower-dimensional space through a probability model while preserving the relative distances between original data points (Van der Maaten and Hinton, 2008). Hyperparameter optimization is crucial for achieving optimal visualization effects. As a core parameter, perplexity, typically ranging from 5 to 50, controls the neighborhood size of each data point, with smaller values revealing local structures and larger values capturing global structures; learning rate affects the compression degree of data points in low-dimensional space, with a typical range of 10-1000, requiring a balance of inter-cluster distances; iteration count, usually between 200-1000, controls algorithm convergence; the early exaggeration parameter, defaulting to 12.0, adjusts the initial inter-cluster distance amplification. This method is primarily used for exploratory data analysis and high-dimensional data visualization, particularly suitable for revealing the essence of complex data structures. Through systematic parameter optimization, it can significantly improve the accuracy and insight of data visualization, providing an effective dimensionality reduction strategy for understanding the structure of complex datasets.

The XGBoost is an implementation of gradient boosting trees that optimizes computation speed and resource efficiency. By constructing multiple layers of decision trees and letting subsequent trees correct the errors of previous ones, XGBoost can efficiently handle large-scale data and provide highly accurate prediction models (Chen and Guestrin, 2016). To achieve optimal performance, hyperparameter optimization is crucial for XGBoost. Learning rate controls the weight of each tree, typically ranging

from 0.01 to 0.3, with smaller values requiring more trees but improving model generalization; maximum tree depth influences the complexity of individual trees, usually between 3-10, with deeper trees prone to overfitting; subsample ratio controls the proportion of data used for training each tree, ranging from 0-1, which helps reduce overfitting; column sampling ratio determines the random feature selection proportion, similarly contributing to model generalization. Through methods such as grid search and random search for parameter tuning, model performance can be significantly enhanced. XGBoost is widely applied in various machine learning competitions and practical problems, favored for its outstanding performance and robust flexibility.

2.5. Feature selection and algorithm optimization

Feature selection is the process of selecting the optimal subset of features from the original feature set, aiming to build more efficient models. In mineral big data research, feature selection plays a crucial role. Mineral data often contains features from various analytical methods, such as chemical compositions, physical properties, and crystal structure parameters, which may exhibit redundancy or strong correlations. Through statistical methods and expert experience, feature selection can identify key parameters that are most indicative of mineral classification and genesis research. This process not only reduces data redundancy, avoids the ‘curse of dimensionality’, and enhances computational efficiency but also identifies important geochemical indicators, providing reliable evidence for geological research and mineral exploration. Additionally, feature selection makes research models easier to understand (Li et al., 2017).

In specific methods of feature selection, we primarily use Recursive Feature Elimination (RFE) (Darst et al., 2018) with Cross-Validation (Browne, 2020). The steps of this method include initially training a model (such as a Support Vector Machine or Decision Tree), evaluating the importance of features, then progressively removing the least important features, and retraining the model after each removal to assess performance. This process is repeated until a preset number of features is reached or model performance no longer improves. Through this method, researchers can effectively select the most valuable features, thereby constructing more precise and interpretable models, providing stronger support for mineral research.

After completing feature selection, it is necessary to optimize the algorithm to further enhance model performance. To ensure that the adjusted hyperparameters are not only effective for a specific data subset but also possess strong generalizability, this study utilized K-Fold Cross-Validation (Anguita et al., 2012) and grid search methods (Bao and Liu, 2006) for hyperparameter tuning. These methods significantly enhance the model's reliability and robustness, enabling it to perform well across multiple independent data subsets, thereby reducing the risk of overfitting. After completing

feature selection, the parameters of the selected algorithm were further adjusted through grid search, systematically exploring different parameter combinations to determine the optimal settings for performance (Liashchynskiy and Liashchynskiy, 2019). This optimization not only improves the model's prediction accuracy but also strengthens its generalization capability. In mineral big data research, appropriate parameter optimization can enhance the model's adaptability to different types of mineral data, thereby achieving more reliable classification or prediction outcomes.

During the process of algorithm tuning and feature selection, we used KNN imputation and CLR transformation methods that had previously performed well in our evaluations. For parameter optimization, we primarily employed grid search and cross-validation. Table 1 displays the parameters optimized through grid search.

For the PCA algorithm, cross-validation and Random Forest were used to assess feature importance. For the t-SNE algorithm, it was integrated with the Recursive Feature Elimination (RFE) and the Random Forest algorithm, comparing the silhouette coefficients of each dimensionality reduction to find the optimal number of features. The LDA algorithm was used cross-validation to find the optimal number of principal components and assessed feature importance with Random Forest, using RFE for feature selection. In the PLS-DA algorithm, RFE and grid search were applied to determine the optimal components and features. For the RF algorithm, parameter optimization was conducted via grid search and selected the best features. In the SVM algorithm, grid search was used for parameter optimization and RFE for feature selection. Finally, for the XGBoost algorithm, parameters optimization was used grid search and the SelectKBest was for feature selection.

3. Results and discussion

3.1. Effect of imputation method on model performance

Box plots show the data after imputation by elements (Fig. 3). For elements like Ag, Cd, Cu, Fe, Mn, and Pb with fewer missing values, the data distributions remain relatively similar across different imputation methods. However, for elements like Co, Ga, Ge, In, Sb, and Sn with more missing values, comparing various imputation methods reveals that KNN, RF, and Mean methods have more significant impacts on data distribution. Among these, KNN has the minimal impact on data distribution, while RF and Mean methods show the largest impact. In most cases, data distributions after Median and Mode imputation appear more stable.

Because only XGBoost algorithm can deal with the blank data, we compared the performance of the XGBoost algorithm on datasets with and without imputed missing values. The results show that the accuracy and F1-score of the model without

imputation are slightly higher than those with imputed values (Table 3). The XGBoost algorithm has an internal mechanism for handling missing values. Its Sparsity-aware feature treats missing values as a special state by default and automatically learns the optimal splitting direction during the training process. This approach, compared to manual imputation, is better at capturing the potential correlation between the missing features and the target variable (Chen and Guestrin, 2016). We used Clr transformation and evaluated five missing value imputation methods using the accuracy and F1-score of LDA and RF algorithms. The results revealed that in the RF model, KNN and Mean methods exhibited the highest overall accuracy, with F1-scores significantly outperforming other methods across different ore deposit types (Table 4). In the LDA model, KNN, Median, Mean, and RF showed similar overall accuracy, with comparable F1-scores across various ore deposit types (Table 5). Synthesizing the evaluation results from both algorithms, KNN was identified as the best imputation method due to its high accuracy and stable performance.

Since the random forest imputation method is often based on complex model predictions, it is prone to introduce bias, excessive noise, and complexity in cases of uneven data distribution or the presence of outliers, which may ultimately lead to a decline in the generalization ability of the algorithm model, resulting in overfitting on the training data. Since the data used is skewed, the Mean imputation method can be affected by extreme values, ultimately introducing bias into the data. The KNN, Mode, and Median imputation methods do not require assumptions about the data distribution and are suitable for various types of datasets, including non-normally distributed data.

3.2. Effect of on data transformation classification algorithms

The distribution pattern of transformed data using different methods was displayed in Fig. 4. The results show that under all transformation methods, the classification performance of t-SNE, LDA, and PCA algorithms when processing RF imputed data was not satisfactory. However, the discriminant plots generated by the LDA algorithm after Clr and Log transformations, excluding RF imputation, could effectively distinguish between SEDEX and MVT types, with a classification accuracy exceeding 0.80, an improvement of about 0.20 compared to other transformation methods (Fig. 5). Additionally, the PCA and t-SNE algorithms also demonstrated better discriminative abilities on data transformed by Clr (Figs. 6 and 7).

The performance of four algorithms, PLS-DA, RF, SVM, XGBoost, were evaluated using multiple metrics including Precision, Recall, F1-score, Support, and Accuracy, with particular emphasis on the results of F1-score and Accuracy (Table. 2). After detailed comparisons, it was found that the PLS-DA algorithm performed best when processing Clr and Log transformed data after KNN and Median imputation.

The SVM algorithm also performed well on data transformed with Clr and Log, particularly excelling with data imputed using the KNN method (Table 2). The XGBoost algorithm and RF algorithm achieved F1-scores and accuracy of almost over 0.95 under most imputation and transformation combinations, showing its stable and robust classification ability in various data processing scenarios.

It is noteworthy that the RF and XGBoost, two widely popular machine learning algorithms, also perform well on datasets without transformations. This finding reveals the high robustness and adaptability of RF and XGBoost when dealing with datasets that do not conform to Clr transformations. Therefore, in certain cases, when the data structure and characteristics are not suitable for complex transformations, these algorithms might be the more ideal choice.

3.3. Effect of feature selection and parameter optimization on classification results

Feature importance was determined using RF, XGBoost, and SVM algorithms. The results show that a similar ranking for twelve elements between these algorithms, with Ge, Mn, Ga, In, and Sn being the most important in the model construction (Fig. 8). For PCA, optimal number of components to be 5, with the first two principal components being the most important as indicated by the eigenvalue plot. After evaluation with the Random Forest algorithm, only five features were selected: Mn, Ga, Ge, Sn, and In. The accuracy rate for each deposit assessment reached 0.89 (Fig. 6, Table 2). For the t-SNE algorithm, the optimal number to be 9, including Fe, Pb, Mn, Co, Ga, Ge, Cd, In, and Sn. The optimization results showed that the clustering of categories for each deposit was tighter, significantly improving the differentiation (Fig. 7). The classification effects evaluated using the Random Forest algorithm reached an accuracy rate of over 0.66 for each deposit (Table 2).

The optimal number of LDA components was 2, and the final selected feature combination included Fe, Mn, Co, Ga, Ge, Ag, Cd, In, Sn, and Sb, achieving the best classification performance. The discriminant plot clearly showed a more distinct classification of MVT and SEDEX deposits (Fig. 5).

In the PLS-DA algorithm, the optimal number of components are 4, with the best feature set including Fe, Pb, Mn, Co, Cu, Ga, Ge, Cd, In, Sn, and Sb. However, the final evaluation results showed F1-score for MVT, Epithermal, and SEDEX types of 0.87, 0.56, and 0.69 respectively, while Skarn and VMS types had F1-score of 0 (Table 2). For the RF algorithm, the optimal feature includes Fe, Mn, Co, Ga, Ge, Cd, In, and Sn. This feature set achieved F1-score of over 0.90 for each deposit classification, with a final accuracy of 0.94 (Table 2). The optimal number of features for SVM algorithm to be 10, including Fe, Pb, Mn, Cu, Ga, Ge, Ag, Cd, In, and Sn. Each deposit F1-score exceeded 0.90, with an accuracy of 0.95 (Table 2). The optimal number of features for

XGBoost algorithm to be 10, including Fe, Pb, Mn, Cu, Ga, Ag, Cd, In, and Sn. This feature set achieved F1-score of over 0.90, with an accuracy of 0.95 (Table 2).

Due to the small amount of data in this study, parameter tuning and feature selection had a significant impact on the final results when using the SVM algorithm. This is because the key to the SVM algorithm is the selection of an appropriate kernel function, which directly affects the performance and classification results of the model. An appropriate kernel function can effectively map data to a higher-dimensional space, thereby improving classification accuracy. Therefore, in this study, meticulous parameter tuning and feature selection for the SVM algorithm significantly enhanced the predictive capability and overall effectiveness of the model.

3.4. Comparisons between different ML methods

Clr-transformed data imputed by KNN were used for a systematic evaluation of seven algorithms. These seven types of algorithms can be divided into supervised and unsupervised algorithms. In unsupervised algorithms, since PCA and t-SNE lack explicit evaluation criteria, besides observing the discriminant plots output by these two algorithms, The random forest algorithm to produce a classification report, assessing the dimension reduction effects of the two algorithms. The results showed that the discriminant plot of PCA was more distinct, and its F1-Score and accuracy are all better than t-SNE (Table 2 and Fig. 7), whereas t-SNE's discriminant plot performed more prominently in optimizing clustering (Fig. 8). This indicates that although PCA is superior in classification effects, t-SNE also holds significant advantages in visualization and clustering, capable of better displaying the local structure of the data.

Among the supervised algorithms, the PLS-DA algorithm performs the worst (Table 5). This may be due to the imbalance in the dataset and the assumption in PLS-DA that there is a linear relationship between the independent variables and the dependent variables. However, by calculating the Pearson correlation coefficients between each trace element and the type of deposit, the results show that most correlation coefficients are close to 0, indicating the absence of linear correlation. The LDA algorithm assumes that the data for each category approximately follows a Gaussian distribution, but the original data was observed to be skewed. Only after using CLR and logarithmic transformations did the classification accuracy improve from 0.60 to about 0.80 (Table 6). The optimized SVM, RF, and XGBoost algorithms demonstrated good robustness (Table 2), highlighting their effectiveness and stability in dealing with complex data. These three algorithms particularly excel in handling data imbalance and nonlinear features, providing more reliable classification results.

4. Conclusion

This study primarily used seven different machine learning algorithms to classify sphalerite data collected from various types of deposits. For skewed distribution data, the KNN, Mode, and Median imputation methods are more suitable to minimize the introduction of unnecessary bias into the data. The PCA, t-SNE, LDA, PLS-DA, and SVM algorithms performed better on data with Clr or Log transformations. The RF and XGBoost algorithms demonstrate good classification capabilities for untransformed data. Despite the limited dataset, feature selection still helps identify key features in the classification process. Through grid search and cross-validation, the algorithm parameters can be optimized. Constrained by the data volume, only SVM significantly improved classification performance by optimizing its kernel function. We emphasized the critical role of data preprocessing and parameter optimization in improving model performance. However, it should be noted that the preprocessing strategies and ML parameters may be different for the dataset with different structures.

CrediT authorship contribution statement

Ying-Hui Gao: Conceptualization, Building algorithm models, Data curation, Visualization, Writing - original draft, Writing – editing. **Xiao-Wen Huang :** Conceptualization, Funding acquisition, Writing - review & editing. **Yu-Miao Meng:** Data curation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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545

546 **Appendix A. Supplementary data**

547 The Python code related to this article can be found online at
 548 <https://github.com/Williamhuizi/deposit-classfication>.

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550 **References**

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Figure captions

Fig. 1 Mineral big data mining or model construction workflow.

Fig. 2 Illustrations showing various machine learning algorithms. (a) Principal component analysis (PCA); (b) t-Distributed Stochastic Neighbor Embedding (t-SNE); (c) Linear Discriminant Analysis (LDA); (d) Partial Least Squares Discriminant Analysis (PLS-DA); (e) Random Forest (RF); (f) Support Vector Machine (SVM); (g) Extreme Gradient Boosting (XGBoost).

Fig. 3 Box and whisker plots of major and trace elements in sphalerite using different data imputation methods. The upper and lower edges of the box represent the 75th and 25th percentiles of the data, respectively. Whiskers indicate the minimum and maximum values. The median is represented by a solid black line, while the mean is shown by an empty circle.

Fig. 4 The Ag data distribution using different transformation methods. All data are imputed by KNN method.

Fig. 5 Deposit type classification diagrams using the LDA algorithm with transformed data by Clr (a), Log (b), Maxabs (c), Robust (d).

Fig. 6 Deposit type classification diagrams using the PCA algorithm with transformed data by Clr (a), Log (b), Maxabs (c), and Robust (d).

Fig. 7 Deposit type classification diagrams using the t-SNE algorithm with transformed data by Clr (a), Log (b), Maxabs (c), and Robust (d).

Fig. 8 Feature importance of different algorithms. (a) RF, (b) SVM, (c) XGBoost.

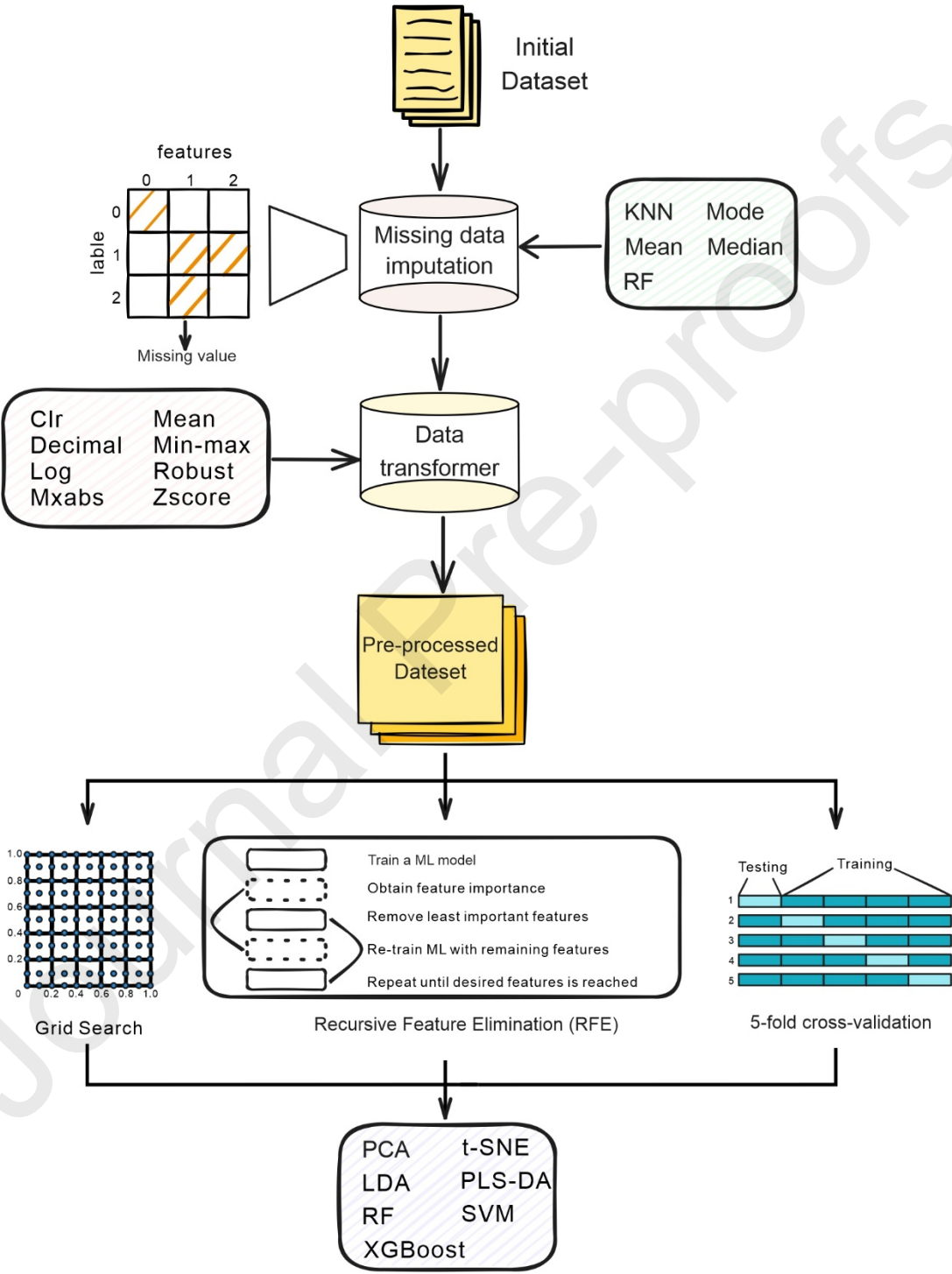


Fig. 1

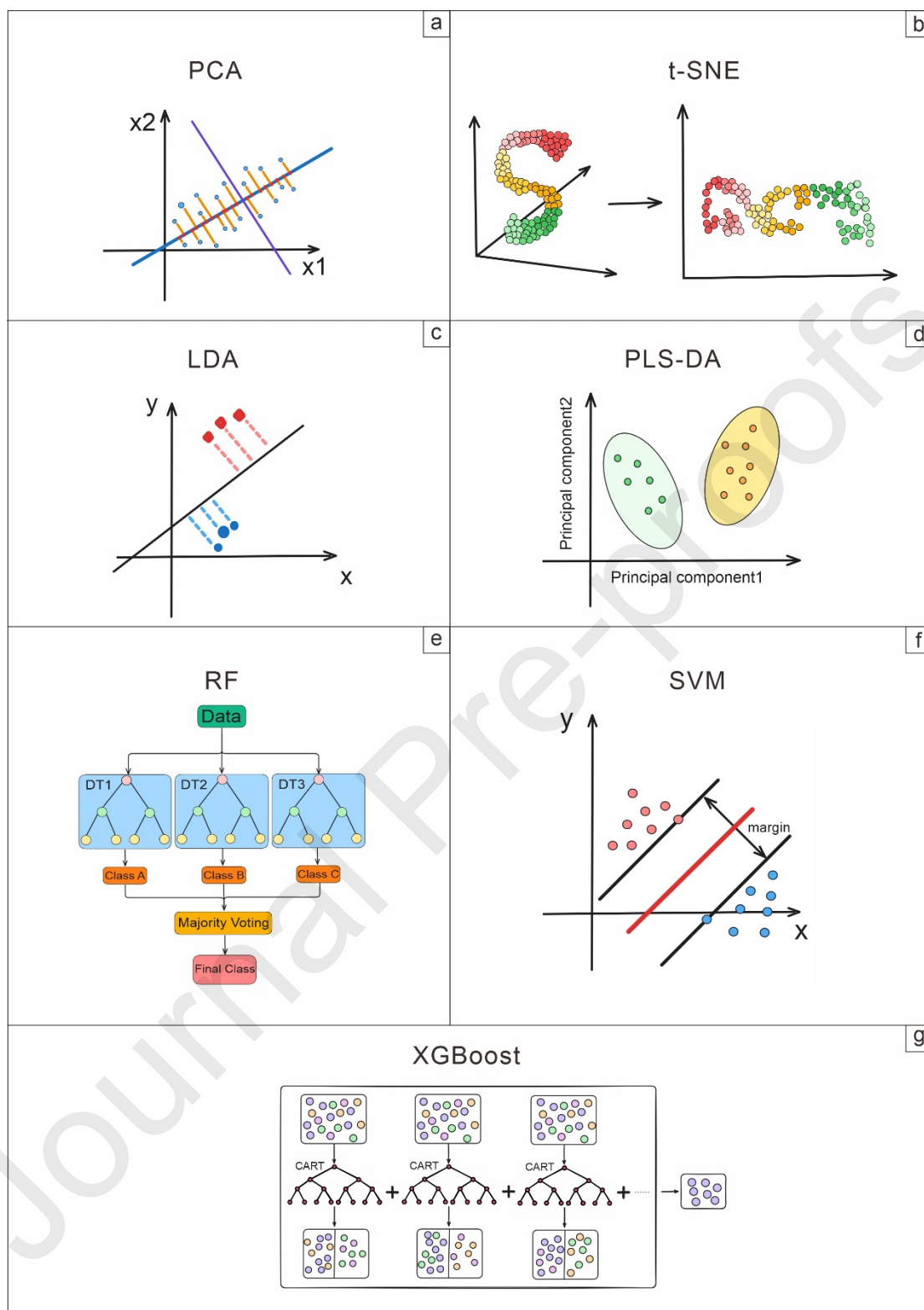


Fig. 2

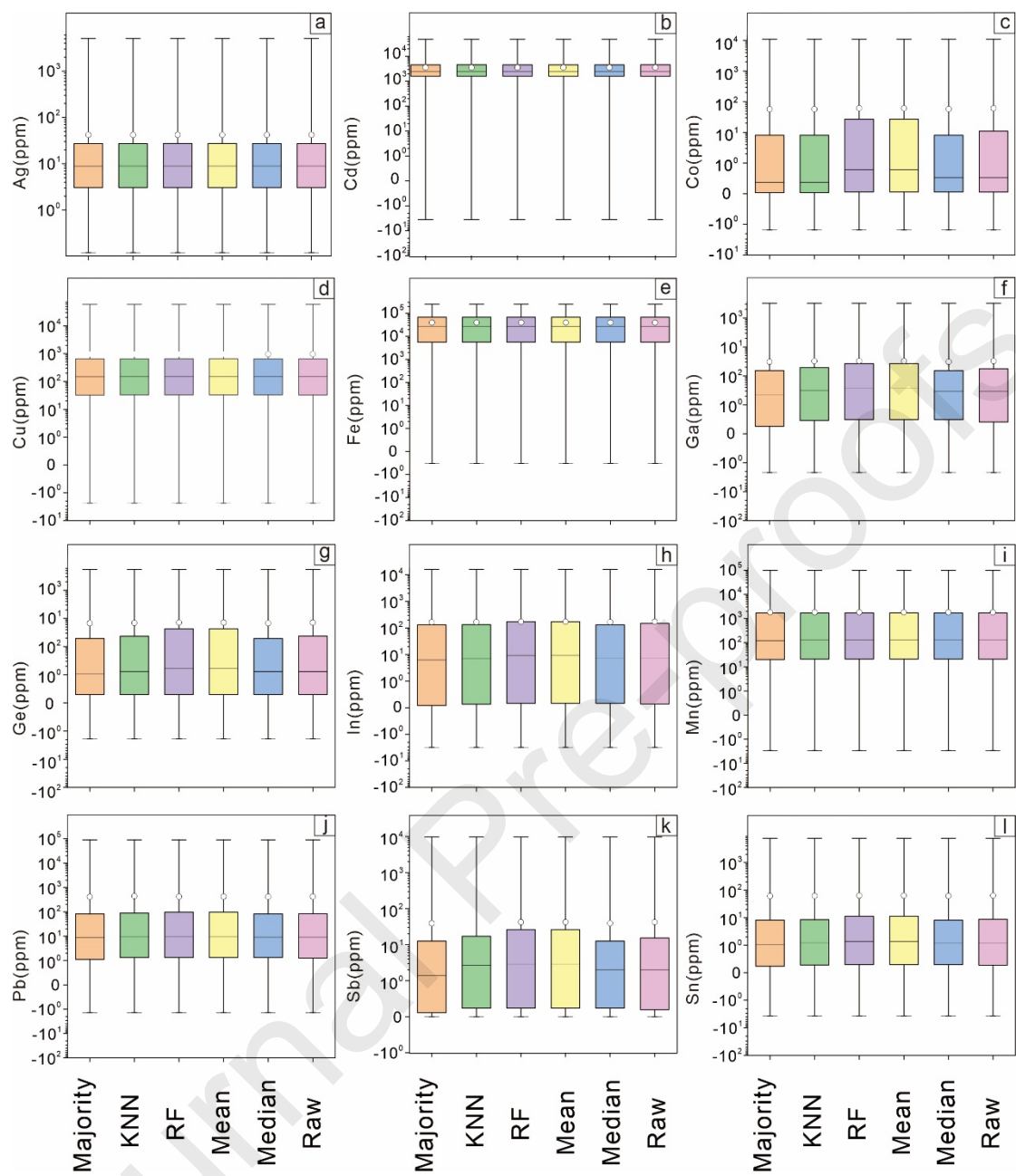


Fig. 3

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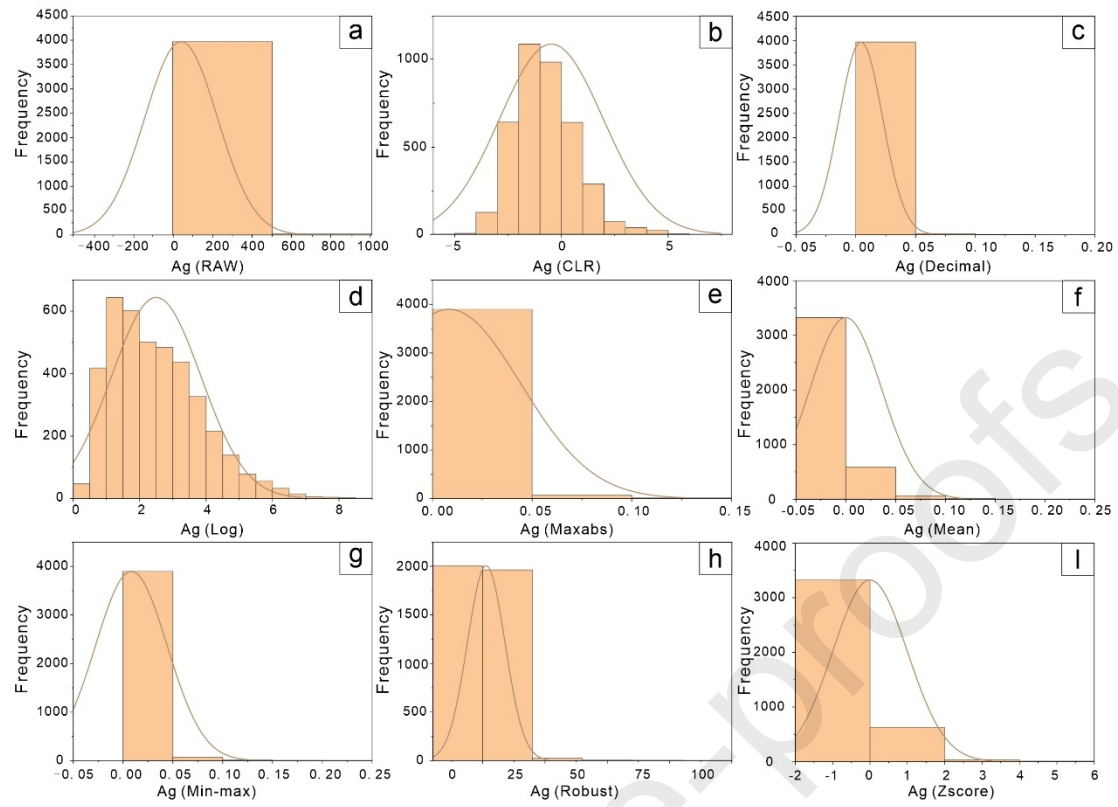


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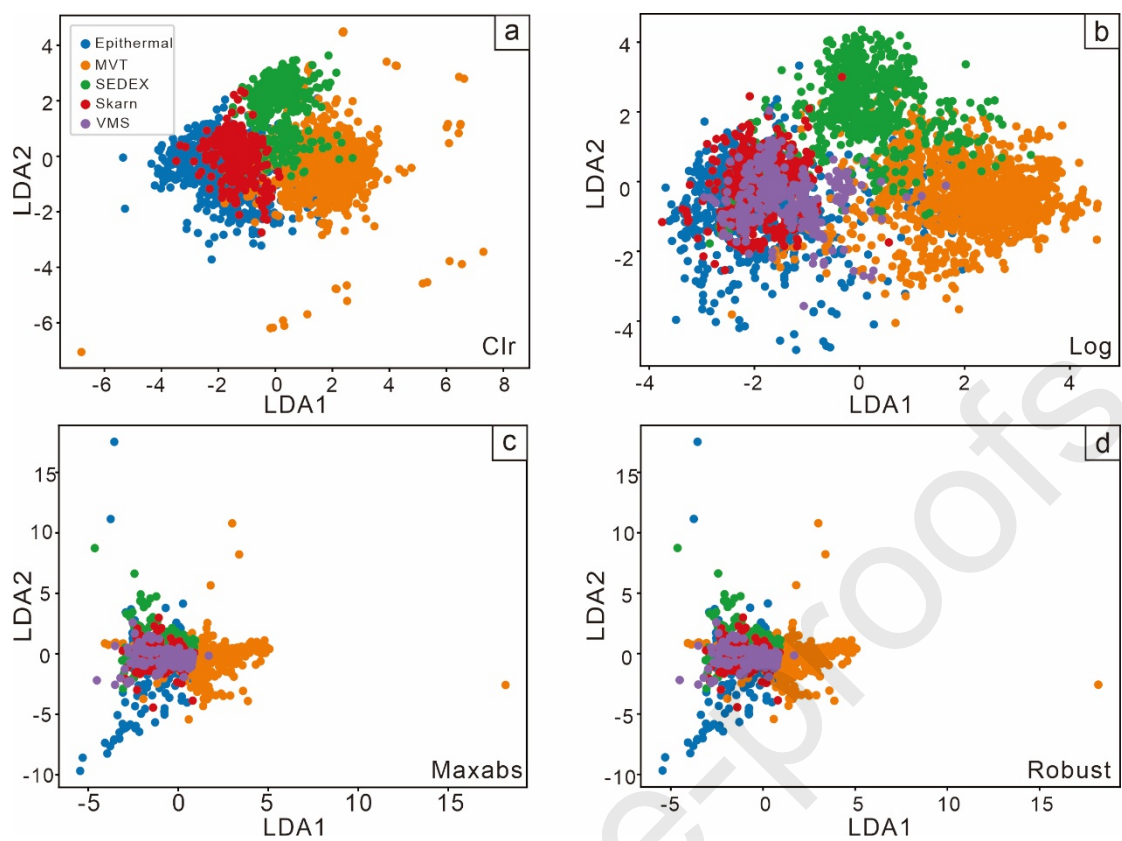


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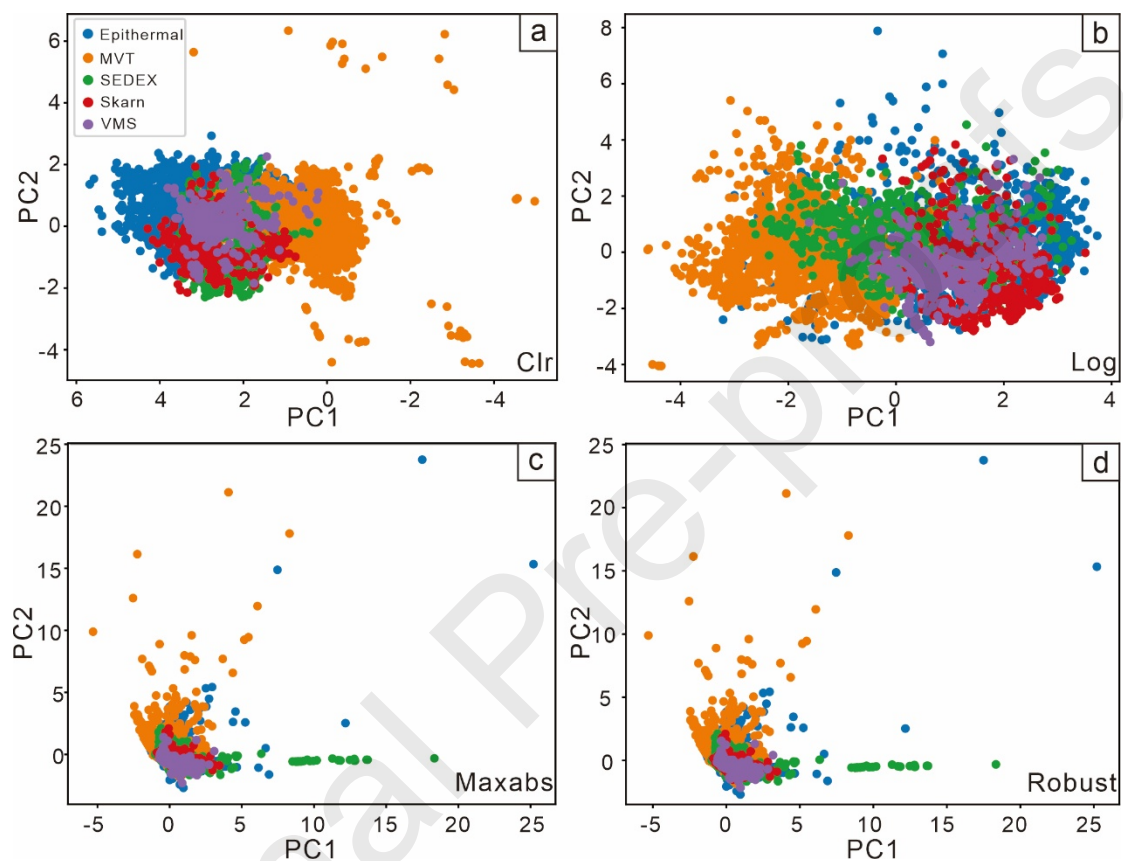


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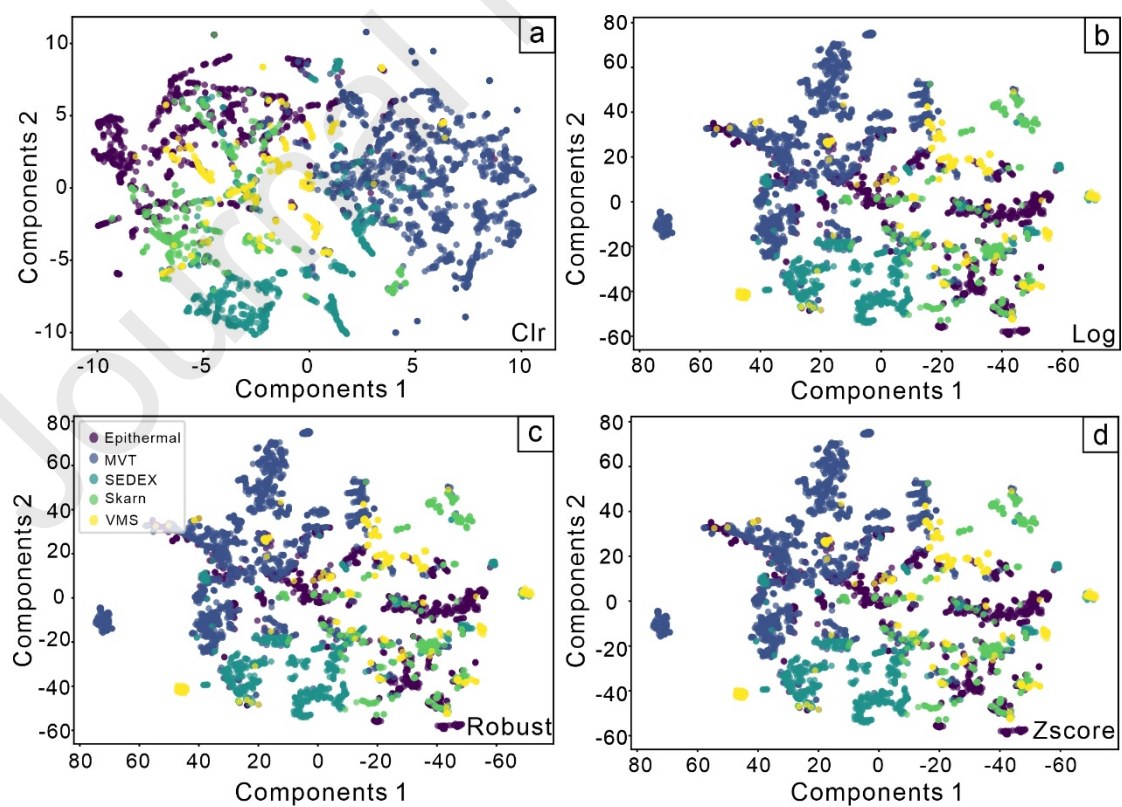


Fig. 7

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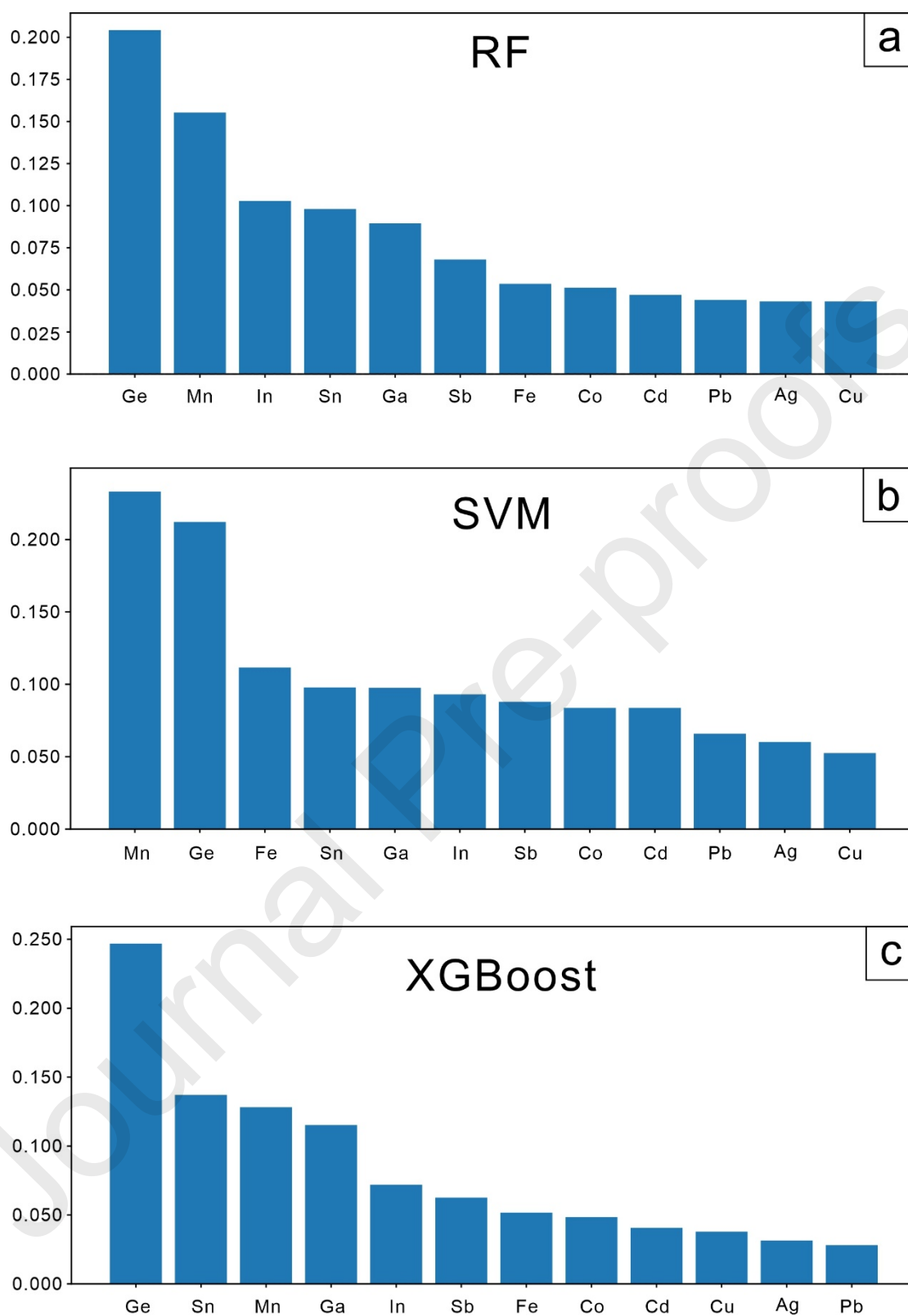


Fig.8

Table 1

Optimal hyperparameters for different machine learning methods.

Base classifier	Hyperparameters	Optimal combinations
PLS-DA	n_features_to_select: [5, 6, 7, 8, 9, 10, 11, 12]	n_features_to_select: 11
	n_components: range(1, max_components + 1)	n_components: 4
	n_estimators: [100, 200]	n_estimators: 100
RF	max_depth: [None, 10, 20]	max_depth: None
	min_samples_split: [2, 5]	min_samples_split: 2
	min_samples_leaf: [1, 2]	min_samples_leaf: 1
	max_features: ['sqrt', 'log2', None]	max_features: 'sqrt'
SVM	C: [0.1, 1, 10, 100]	C: 100
	kernel: ['linear', 'rbf']	kernel: 'rbf'
	gamma: ['scale', 'auto']	gamma: 'auto'
XGBoost	n_estimators: [50, 100, 200]	n_estimators: 200
	max_depth: [3, 4, 5]	max_depth: 5
	learning_rate: [0.01, 0.1, 0.2]	learning_rate: 0.2

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851 **Table 2**

F1-score and accuracy of different algorithms or assemblages of algorithms after feature selection and parameter optimization. and parameter optimization.

Algorithms	F1-Score					Accuracy	
	Epithermal	MVT	SEDEX	Skarn	VMS		
LDA	0.75	0.90	0.82	0.64	0.63	0.80	
PLS-DA	0.56	0.87	0.69	0.00	0.00	0.63	
RF	0.92	0.97	0.95	0.92	0.91	0.94	
SVM	0.94	0.97	0.96	0.92	0.92	0.95	
XGBoost	0.95	0.98	0.96	0.91	0.92	0.95	
PCA+RF	0.87	0.95	0.93	0.80	0.82	0.89	
t-SNE+RF	0.67	0.85	0.48	0.38	0.40	0.66	

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Table 3

F1-score and accuracy of the XGBoost algorithm using imputed and unimputed data

Algorithm	Data set	F1-Score					Accuracy
		Epithermal	MVT	SEDEX	Skarn	VMS	
XGBoosst	Imputed	0.94	0.97	0.97	0.92	0.92	0.95
	Unimputed	0.96	0.99	0.99	0.97	0.97	0.98

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922 **Table 4**

923 F1-score and accuracy of the random forest algorithm using different imputation
 924 methods.

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Methods	F1-Score					Accuracy
	Epithermal	MVT	SEDEX	Skarn	VMS	
KNN	0.93	0.95	0.95	0.94	0.95	0.95
Median	0.93	0.97	0.94	0.90	0.90	0.94
Mean	0.93	0.97	0.97	0.93	0.95	0.95
Mode	0.93	0.97	0.95	0.91	0.92	0.95
RF	0.94	0.97	0.97	0.94	0.97	0.96

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959 **Table 5**

960 F1-score and accuracy of the LDA algorithm using different imputation methods.

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Method	F1-Score					Accuracy
	Epithermal	MVT	SEDEX	Skarn	VMS	
KNN	0.75	0.89	0.82	0.62	0.58	0.79
Median	0.76	0.88	0.81	0.63	0.60	0.79
Mean	0.75	0.89	0.81	0.63	0.57	0.79
Mode	0.74	0.90	0.82	0.61	0.56	0.78
RF	0.76	0.88	0.81	0.63	0.60	0.79

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997 **Table 6**

998 F1-score and support of the PLS-DA algorithm.

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Deposit type	F1-score	Support
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Epithermal	0.56	982
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MVT	0.87	1456
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SEDEX	0.69	714
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Skarn	0.00	436
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VMS	0.00	415
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- 1041 ➤ Feature selection can optimize algorithm and enhance the interpretability of results.
- 1042 ➤ Grid search and cross-validation can effectively enhance the accuracy of the
- 1043 algorithm.
- 1044 ➤ XGBoost demonstrates superior classification performance on unfilled data.
- 1045